SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5112

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Nirmal Kasi 192424248 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Nirmal Kasi

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:

   a. Show how each attack leads to specific threats and risks
   b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:

   a. Show differences and similarities
      i. Map how each affects the Data structure and Security level
      ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

   a. Show relationships between these concepts
   b. Analyze how each contributes to secure communication
   c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

   a. Show how each concept is mathematically connected
   b. Map their role in encryption/decryption
   c. Analyze which concept is most critical for public key cryptography and justify

1. Concept Map: Security Attacks, Threats, Risks and Security Goals (CIA)

```
                         SECURITY ATTACKS
                                |
     ┌──────────────────┬──────────────────┬──────────────────┐
     |                  |                  |                  |
Passive Attack    Active Attack     Malware Attack     Insider Attack
     |                  |                  |                  |
     ▼                  ▼                  ▼                  ▼
Information       Data Tampering     System Damage      Data Theft
Disclosure        Unauthorized       Service            Privilege Abuse
                  Modification       Disruption
     |                  |                  |                  |
     └──────────────────┴──────────────────┴──────────────────┘
                               |
                            THREATS
                               |
           Unauthorized Access • Data Loss • Financial Loss
                               |
                               ▼
                             RISKS
                               |
     ┌──────────────────┬───────────────┬───────────────────┐
     |                  |               |
Loss of Privacy    Business Loss    Reputation Damage
                               |
                               ▼
                      SECURITY GOALS (CIA)
                               |
   ┌─────────────────┬─────────────────┬─────────────────────┐
   |                 |                 |
Confidentiality  Integrity         Availability
   |                 |                 |
Encryption       Hashing            Backups
Access Control   Digital Signatures   Redundancy
```

Real-world examples

Attack	Threat	Risk	Security Goal
Phishing	Credential theft	Bank account compromise	Confidentiality
SQL Injection	Database modification	Wrong customer records	Integrity
DDoS Attack	Server unavailable	Website downtime	Availability
Insider Attack	Data leakage	Reputation loss	Confidentiality + Integrity

Analysis

- **Confidentiality** prevents unauthorized disclosure of information.
- **Integrity** ensures information is not modified illegally.
- **Availability** ensures systems remain accessible.
- Together these three goals reduce the impact of attacks and risks.

2. Concept Map: Comparison of Substitution Techniques

```
                    SUBSTITUTION TECHNIQUES
                              |
```

```
┌──────────────┬──────────────┬──────────────┬──────────────┐
│              │              │              │
Caesar     Monoalphabetic  Vigenère      Hill Cipher
│              │              │              │
Single Key   One Fixed      Keyword       Matrix Key
Shift         Alphabet       Repeats       (Linear Algebra)
│              │              │              │
Low           Medium         High          Very High
Complexity    Complexity     Complexity    Complexity
│              │              │              │
Weak          Moderate       Strong        Strongest
Security      Security       Security      Security
```

Comparison

Cipher	Key Usage	Complexity	Security Strength
Caesar	Single shift	Very Low	Weak
Monoalphabetic	Fixed substitution alphabet	Medium	Moderate
Vigenère	Repeating keyword	High	Strong
Hill Cipher	Matrix key	Very High	Very Strong

Most Secure

Hill Cipher

Reasons:

- Uses matrix multiplication.
- Encrypts multiple letters together.
- Harder to perform frequency analysis.
- Larger key space than other substitution techniques.

3. Concept Map: Classical Encryption Schemes

```
              CLASSICAL ENCRYPTION
                      │
      ┌───────────────┴──────────────┐
      │                              │
Substitution Techniques      Transposition Techniques
      │                              │
Replace Letters             Rearrange Letters
      │                              │
Character Changes         Position Changes
      │                              │
Meaning Hidden            Order Hidden
      │                              │
Examples                  Examples
Caesar                    Rail Fence
Vigenère                  Columnar
Hill Cipher               Route Cipher
```

Similarities

- Both protect plaintext.
- Both require keys.
- Both are classical encryption methods.

Differences

Substitution	Transposition
Changes characters	Rearranges characters
Frequency changes	Frequency remains same
Better protection	Easier to analyze

Which is more vulnerable?

Transposition Cipher

Reasons:

- Letter frequencies remain unchanged.
- Easier to break using frequency analysis.
- Small key space.
- Brute-force attacks are simple.

4. Concept Map: Steganography

```
                    STEGANOGRAPHY
                          │
        ┌─────────────────┼─────────────────┐
        │                 │                 │
  Data Hiding         Watermarking      Survivability
        │                 │                 │
Hide Secret         Ownership        Resist Attacks
Information          Protection        and Compression
        │                 │                 │
LSB Method          Copyright        Robust Embedding
        │                 │                 │
Secure Transfer     Authentication   Reliable Recovery
```

Relationship

- Data Hiding conceals confidential information.
- Watermarking proves ownership.
- Survivability ensures hidden data remains after modifications.

Contribution

Concept	Contribution
Data Hiding	Protects secret communication
Watermarking	Prevents copyright theft
Survivability	Ensures hidden data survives image processing

Real-world applications

Concept	Application
Data Hiding	Military communication
Watermarking	Copyright protection for digital images
Survivability	Medical image authentication

5. Concept Map: Number Theory in Cryptography

```
                    NUMBER THEORY
                          |
          ----------------------------------
          |                                |
Extended Euclid Algorithm        Euler Phi Function
          |                                |
Finds GCD                       Counts Coprimes
          |                                |
Computes Modular Inverse         Euler's Theorem
          |                                |
          ----------------------------------
                          |
            Fermat's Little Theorem
                          |
         Special Case of Euler's Theorem
                          |
                          ▼
            RSA Public Key Cryptography
```

Mathematical Connection

- Extended Euclidean Algorithm computes the **Modular Inverse**.
- Modular Inverse is required for RSA decryption.
- Euler's Phi Function computes $\varphi(n)$.
- Euler's Theorem uses $\varphi(n)$.
- Fermat's Little Theorem is a special case of Euler's Theorem for prime numbers.

Role in Encryption/Decryption

Concept	Role
Extended Euclid	Finds private key
Modular Inverse	RSA decryption
Euler Phi Function	Key generation
Euler's Theorem	RSA correctness
Fermat's Little Theorem	Efficient modular exponentiation

Most Important Concept

Euler's Phi Function ($\varphi(n)$)

Justification

- RSA key generation depends directly on $\varphi(n)$.
- It is used to calculate both public and private keys.
- Without Euler's Phi Function, RSA encryption and decryption cannot be performed correctly.

- It forms the mathematical foundation of modern public-key cryptography.

Conclusion

These concept maps illustrate the relationships among classical cryptography techniques, security principles, steganography, and number theory. They also explain how each concept contributes to secure communication and why modern cryptographic systems rely on strong mathematical foundations rather than simple classical ciphers.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation